

Gabriel Guzman Blanco

Denver, Colorado

✉ gabeguzman08@gmail.com ☎ (720) 512-9752 🌐 github.com/plantmandev 🔗 linkedin.com/in/gabeguzman

PROFESSIONAL EXPERIENCE

Lab Technician

Wagner Equipment Co. Arvada, CO
Aug 2025 - Present

- Analyze **10,000+** oil samples from heavy equipment transmissions, hydraulics, final drives, and engines to assess contamination levels, oil degradation, and component wear patterns
- Conduct multi-instrument diagnostic testing to identify critical issues including coolant contamination, water ingress, fuel dilution, oxidation, and abnormal wear metals in construction and mining equipment
- Generate detailed technical reports for preventative maintenance programs, enabling clients to avoid costly equipment failures and extend component service life
- Perform Quality Assurance (QA) / Quality Control (QC) procedures including instrument calibration, reference standards validation, and method verification to ensure regulatory compliance and data accuracy
- Process samples using **ICP-OES** for elemental analysis, **FTIR spectroscopy** for chemical composition, **ion chromatography** for fuel contaminants, and **viscometry** for oil condition assessment

Aquatic Toxicologist

SeaCrest Group Louisville, CO
Aug 2025 - Oct 2025

- Conducted acute and chronic **Whole Effluent Toxicity (WET)** tests using cladoceran cultures in compliance with **U.S. Environmental Protection Agency (EPA)** protocols
- Maintained in-house cultures of *Ceriodaphnia dubia*, *Daphnia pulex*, and *Daphnia magna* (small freshwater crustaceans used as standard aquatic test organisms) for routine toxicological testing, supporting **15–20 WET** tests weekly
- Analyzed incoming client samples to identify and quantify environmental contaminants
- Performed **QA** procedures, including reference toxicant testing to validate culture health and test reliability
- Calibrated and maintained laboratory equipment to ensure accuracy and **EPA** compliance

Research Assistant

Gettysburg College Gettysburg, PA
Aug 2020 - Dec 2023

- Supported **25+** systematic sampling campaigns and managed data collection protocols for multi-site ecological research project
- Executed field sampling protocols across **15** study sites, tracking population dynamics and environmental correlations using statistical analysis
- Compiled quarterly analytical reports for **National Park Service** stakeholders, presenting data-driven recommendations for **1,500+ acres** of managed habitat
- Performed statistical analysis using **Python** and **R** for temporal data processing, correlation studies, and data visualization
- Presented research findings at **3** scientific conferences, demonstrating technical communication skills

SKILLS

Data Analysis & GIS	Python (pandas, numpy, seaborn), R (tidyverse, ggplot2), SQL , PostGIS , QGIS
Field Methods	Environmental sampling, biodiversity assessments, continuous water quality monitoring, FAA Part 107 certified

EDUCATION

B.S. in Environmental Studies

Gettysburg College Gettysburg, PA

Concentrations: **Environmental Science**, Wildlife and Conservation Ecology & GIS and Spatial Analysis

Relevant Coursework: General Chemistry, Introduction to Scientific Computation, **Applied Statistics**

Minor: Biology

PROJECTS

Dam Removal Impact Assessment

- Led **3-person** analytical team investigating environmental impacts on **261 acres** using systematic sampling and sediment coring protocols
- Performed **Loss on Ignition (LOI)** combustion analysis achieving **24.94%** organic content, **6.2x** higher than regional baselines
- Analyzed **7,300+** water quality data points (2003–2023) using **Python** for statistical analysis of pH, dissolved oxygen, turbidity, and nutrients
- Operated **YSI6600V2** data loggers and **Hydrolab** sondes for continuous water quality monitoring

Butterfly Population Analysis

- Conducted **25+** systematic field surveys using transect sampling methods across **40+** species
- Performed taxonomic identification using dichotomous keys and photographic documentation protocols